

VALIDATION REPORT

OPERATIONAL QUALIFICATION

RELIABILITY DEMONSTRATION TESTING MODULE

JR Validated Environment — Reliability Demonstration Testing Module v2.0

Field	Value
Document Number	JR-VR-RDT-001
Title	Validation Report — Operational Qualification, Reliability Demonstration Testing Module
System	JR Validated Environment — RDT Module
Module Version	2.0
Document Version	3.0
Validation Plan	JR-VP-RDT-001 v2.0
OQ Result	PASS — 31 / 31 tests passed, 0 failures, 0 deviations
Evidence File	rdt_oq_execution_20261001T171420.txt
Execution Date	2026-10-01
Host	Joeps-MacBook-Air.local
R Version	4.6.0
Python Version	3.11.9
pytest Version	pytest 8.3.4
Status	Approved
Author	Joep Rous
Reviewer	[Name]
Approver	[Name]

1. Purpose and Scope

This document records the results of the Operational Qualification (OQ) for the Reliability Demonstration Testing (RDT) module of the JR Validated Environment. It is produced by executing the automated test suite defined in JR-VP-RDT-001 and provides evidence that both RDT scripts perform correctly for all specified user requirements.

Evidence source: rdt_oq_execution_20261001T171420.txt

Execution timestamp: 2026-10-01T17:14:20

2. Test Environment

Component	Value
Host	Joeps-MacBook-Air.local
R Version	4.6.0
Python Version	3.11.9
pytest Version	pytest 8.3.4
Test runner	repos/rdt/admin_rdt_oq
Test files	repos/rdt/oq/test_rdt_plan.py, test_rdt_verify.py
Evidence file	rdt_oq_execution_20261001T171420.txt

3. Overall OQ Result

Script	Tests	Passed	Failed	Result
jrc_rdt_plan	13	13	0	PASS
jrc_rdt_verify	18	18	0	PASS
Total	31	31	0	PASS

OQ Verdict: PASS

4. Test Results — jrc_rdt_plan

TC ID	Description	Result
TC-RDT-PLAN-001	Bogey mode: exit 0, 'Binomial' in output	PASS
TC-RDT-PLAN-002	Weibayes mode: exit 0, 'Weibayes' in output	PASS
TC-RDT-PLAN-003	Bogey k=0: n=45 (ref: ceil(log(0.10)/log(0.95))), pure Python)	PASS
TC-RDT-PLAN-004	Weibayes k=0 AF=1 beta=2: n=45 (AF ² =1, beta-independent)	PASS
TC-RDT-PLAN-005	Weibayes k=0 beta=2 AF=2: n=12 (ref: ceil(log(0.10)/(log(0.95)*4)))	PASS
TC-RDT-PLAN-006	k_allowed=1: n=77 in output (exact binomial); '<- plan' marker on k=1 row	PASS
TC-RDT-PLAN-007	--help: exit 0, 'Usage' in output	PASS
TC-RDT-PLAN-008	Beta sensitivity table shown when accel_factor > 1	PASS
TC-RDT-PLAN-009	PNG *_jrc_rdt_plan.png written to ~/Downloads/	PASS
TC-RDT-PLAN-010	--reliability=1.5: non-zero exit	PASS
TC-RDT-PLAN-011	--confidence=0: non-zero exit	PASS
TC-RDT-PLAN-012	Missing --confidence: non-zero exit	PASS
TC-RDT-PLAN-013	Bypass protection: direct Rscript without RENV_PATHS_ROOT	PASS

5. Test Results — jrc_rdt_verify

TC ID	Description	Result
TC-RDT-VER-001	Happy path: exit 0, 'Verdict' in output	PASS
TC-RDT-VER-002	PASS verdict n=45 k=0; R_lower≈0.9501 ±0.0002 (exact Beta(1,n) ref)	PASS
TC-RDT-VER-003	FAIL verdict n=20 k=0; R_lower≈0.8913 ±0.0002 (exact Beta(1,n) ref)	PASS
TC-RDT-VER-004	Weibayes PASS n=45 beta=2; R_demo≈0.9501 ±0.0002 (pure Python ref)	PASS
TC-RDT-VER-005	PNG *_jrc_rdt_verify.png written to ~/Downloads/	PASS
TC-RDT-VER-006	All units failed early: R_lower < 0.01 (edge case n-k=0)	PASS
TC-RDT-VER-007	n=50 k=0 R_claim=0.90: R_lower in [0.953, 0.959] (pure Python ref)	PASS
TC-RDT-VER-008	No arguments: exit 0, 'Usage' in output	PASS
TC-RDT-VER-009	File not found: non-zero exit	PASS
TC-RDT-VER-010	Missing column 'time': non-zero exit, 'time' in error	PASS
TC-RDT-VER-011	Non-numeric time values: non-zero exit	PASS
TC-RDT-VER-012	Bypass protection: direct Rscript without RENV_PATHS_ROOT	PASS
TC-RDT-VER-013	--report → *_rdt_verification_report.docx or .html in ~/Downloads/	PASS
TC-RDT-VER-014	--report JSON sidecar *_rdt_verification_report_data.json (or .docx accepted)	PASS
TC-RDT-VER-015	--report JSON: report_type=="rdt", verdict_pass is True or False	PASS
TC-RDT-VER-016	Plan → verify round trip: planned n PASSES with k=1, n - 1 FAILS	PASS
TC-RDT-VER-017	--report with "" in the file name: valid JSON / .docx produced	PASS
TC-RDT-VER-018	Late failures in Weibayes df: 40 units, 2 failed at 6000 → R_demo 0.9117 FAIL	PASS

6. Deviations

No deviations. All test cases passed.

7. Conclusion

The Operational Qualification of the JR Validated Environment RDT module v2.0 is complete. All 31 automated test cases defined in JR-VP-RDT-001 v2.0 have been executed and passed on 2026-10-01. The module is qualified for use in FDA design verification activities under 21 CFR 820.30(f) and ISO 13485:2016 §7.3.6.

8. Approval Signatures

Role	Name	Signature	Date
Author	Joep Rous		2026-10-01
Reviewer	[Name]		
Approver (QA)	[Name]		